

Chandan Kumar Singh

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Summary

Software Engineer with 2 years of experience building scalable, distributed microservices (Java, Spring Boot, Kafka, AWS) for banking and e-commerce platforms; reduced API latency by 40–50% via caching and messaging optimisations; strong DSA foundation (1000+ problems solved) with CodeChef Global Rank 14.

Experience

Tata Consultancy Services

Jan 2025 – Present

Software Engineer — Java, Spring Boot, Microservices

- Designed and owned RESTful APIs for business-critical banking workflows, reducing average response latency by 40% through service-layer optimizations (caching, query tuning, async processing).
- Architected a centralised API Gateway with secret-key authentication and refresh-token rotation, cutting inter-service authentication overhead by 60% and improving distributed system reliability.
- Built secure integrations with external banking systems using AES-256 encryption and defensive controls against MITM, Replay, and Request Tampering attacks, with no reported security incidents in production.
- Owned end-to-end delivery of internal automation tooling, eliminating 64% of manual operational effort and improving release velocity.
- Developed a bulk upload pipeline for concurrent ingestion of thousands of records per request, improving throughput by 35% with zero data-consistency issues.

Projects

Clothify Shopping Application | [GitHub](#)

Java, Spring Boot, Hibernate, PostgreSQL, Redis, Kafka, Docker, Kubernetes, AWS

- Designed and built an e-commerce platform using microservices architecture following SOLID principles, with clean RESTful API contracts enabling independent development and deployment across services.
- Implemented the SAGA pattern with event-driven architecture (Kafka) to manage distributed transactions across Order, Payment, and Inventory services — ensuring data consistency via compensating transactions on failure (e.g., automatic inventory rollback if payment fails).
- Engineered fault-tolerant inter-service communication using Feign Client with the Circuit Breaker pattern, preventing cascading failures during service outages and improving system resilience.
- Optimized performance by integrating Redis for in-memory caching to reduce response latency, with Kafka enabling asynchronous, decoupled communication between services.
- Containerized with Docker and orchestrated via Kubernetes; deployed on AWS with API Gateway and Load Balancing for scalability under variable traffic.

Technical Skills

Languages: Java, Python, SQL, C++, JavaScript, TypeScript

Frameworks: Spring Boot, Spring Security, Hibernate, Kafka, REST API, Docker, Kubernetes, AWS (EC2, S3), FastAPI, React.js

Databases: MySQL, PostgreSQL, Redis, FAISS, ChromaDB

Cloud: AWS (EC2, S3), Docker, Kubernetes, Jenkins

Tools: Git, GitHub, SVN, IntelliJ IDEA, Postman

AI/LLM Engineering: LangChain, RAG Pipelines, Vector Search, Prompt Engineering, Claude CLI, Agent

Achievements

- Solved 1000+ algorithmic problems across LeetCode and GeeksForGeeks; strong depth in Graphs, Trees, Dynamic Programming, Greedy, and Backtracking.
- Global Rank 14 in CodeChef Starters 67 among 10,000+ participants — top 0.014% globally.

- Founded and scaled ZERO ONE Coding Club to 1,000+ active members, mentoring in competitive programming and system design.

Certifications

- Claude Certified Developer - Foundations
- Microsoft Certified: Azure AI Engineer Associate

Education

Motihari College of Engineering, Motihari

B.Tech in Computer Science and Engineering (CSE) | CGPA: 9.0 / 10

Oct 2020 – Aug 2024